

Assignment			
Aspect	Conceptual	Logical	Physical
French name	modèle conceptuel des Données (MCD)	Modèle logique des Données (MLD)	Modèle physique des Données (MPD)
Meaning	High-level business view of data & process	structured design independent of DBMS	Actual implementation in a specific system
Question answered	WHAT?	How?	WITH WHAT?
Tech. used	business concepts only	technology neutral structure	DBMS, SQL, programming
Data model used	Conceptual data model	logical data model	physical data model
Process model used	Conceptual process model	logical process model	physical process model
Audience	Manager, client analysts	Designers, system architects	Developers, DBAs
Output	Entities, attributes, relationships	Tables, PKs, FKs	SQL tables, code

2) Bank account System

Level: Conceptual (WHAT?)

Focus: Business understanding

Data (CDM):

- Entity: Account (Account ID, Balance, Type)
- Entity: Customer (Customer ID, Name, Address)
- Relationship: Customer owns account

Process (CPM):

- open Account
- Deposit Money
- withdraw money

Level 2: logical (HOW?)

Focus: structured design

* Data (LDM):

- Customer (Customer ID PK, Name, Address)
- Account (Account ID PK, Balance, Type, Customer FK)

* process (LPM):

- Define steps for deposit:
 1. check account
 2. update balance
 3. record transaction

Level 3: physical (WITH WHAT?)

Focus: implementation

Level 3: physical
(with what?)
SQL, DBMS, code

Level 2: logical
(HOW?)
Tables, PKs, FKs

Level 1: conceptual
(WHAT?)
Entities, Relationships

